



Collaboration patterns and patenting: Exploring gender distinctions



Yu Meng*

Public Policy and Global Affairs Programme, Nanyang Technological University & Longus Institute for Development and Strategy, Singapore

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ABSTRACT

Drawing upon the theory of gender frame, the research on gender in science, and social network studies, this paper focuses on the social mechanism of collaboration, specifically the *boundary-spanning collaboration*, to understand the gender gap in academic patenting in the U.S. Correspondingly, the author developed a few hypotheses for empirical testing. The results show that, else being equal, only collaboration with industry would significantly increase the probability of patenting for female academic scientists, but this helps explain considerable difference in patenting between female and male academics. The findings are discussed along with the limitations and policy implications at the end.

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1. Introduction

Both female doctoral degree recipients in science and engineering (S&E) and female professionals in scientific workplace have increased steadily in recent decades (Rossiter, 1995; NSF, 2008), but research has continuously found sexism against women in science (Editorial, 2013). The persistent and pervasive gender inequalities in science are detrimental to social values and economic development by reinforcing negative stereotypes and discouraging talented people from participating in and contributing to science (Hanson, 1996; Fox, 2008). Economically, it is estimated by Hunt et al. (2012) that closing the gap between female and male S&E degree holders would increase gross domestic product (GDP) per capita by 2.7% in the U.S. Other socially harmful consequences include the loss of talents and the exclusion of the specific types of knowledge women develop and maintain (Schiebinger, 2008; Kugele, 2010). Hence, the gender gap in science is deemed an ongoing focus of scholarly research and policy intervention.

Research productivity, because of its central role in the success of a scientific career, has attracted the most attention (Fox and Stephan, 2001). While vast previous research on gender and productivity focused on publication records and documented a pattern favoring male scientists (see Cole and Zuckerman, 1984; Xie and Shauman, 1998; Pripic, 2002 for literature review),

concerns have been increasingly cast on another indication of research productivity, *academic entrepreneurship*¹ (Ding and Choi, 2011; Franzoni and Lissoni, 2009) or academic research targeting commercial returns through patents, licenses, products, and many other forms of outcomes. Since commercial involvement is assumed to closely related to established status and a high level of publication productivity (Zucker and Darby, 1996; Stuart and Ding, 2006) and women are generally less successful on these aspects, scholars worry that the change toward commercialization would reinforce women's disadvantages in the profession of science (Whittington and Smith-Doerr, 2005, 2008; Ding et al., 2006; Murray and Graham, 2007; Haeussler and Colyvas, 2011).

Correspondingly, scholars have set out to examine women's status in various forms of academic entrepreneurship, from disclosure (Duque et al., 2005), licensing (Thursby and Thursby, 2005), serving in a firm's scientific advisory board or SAB (Stephan and El-Ganainy, 2007), patenting (e.g., Whittington and Smith-Doerr, 2005; Frietsch et al., 2009; Meng and Shapira, 2010; Blume-Kohout, 2014), to firm founding (Ding, 2004). These efforts have resulted in a consistent finding of a gender gap to women's disadvantage, but have not yet reached to the consensus regarding the underlying explanations. Related to the ambiguity is insufficient attention to collaboration, a process generally assumed to promote academics' research and commercial productivity (see Bozeman et al., 2013

¹ Some scholars use this concept to refer only to firm founding (e.g. Ding and Choi, 2011), but I adopted a broader definition that refers to various commercial activities including patenting, consulting, research collaboration with industry, and firm formation (Franzoni and Lissoni, 2009).

* Tel.: +65 91722579.

E-mail address: mengyu2211@hotmail.com